

Muon vs. Adam

Element-wise (Adam) vs. orthogonalized (Muon) vs. Kronecker-curvature (Shampoo)

Why DeepSeek-V4 trains its weight matrices with Muon and keeps AdamW only for the rest

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The one-sentence difference. Adam treats a weight matrix as a *bag of independent scalars* and gives each its own adaptive step size; Muon treats it as a *linear operator* and reshapes the whole update so every singular direction gets an equal-sized step. Adam conditions *per coordinate*; Muon conditions *per spectrum*.

1 Adam / AdamW — per-coordinate adaptivity

the robust default

Adam keeps two running averages *per parameter*: the gradient (first moment m) and its squared magnitude (second moment v). The step divides the smoothed gradient by its own RMS, so noisy or large-gradient coordinates take smaller steps:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t \odot g_t, \\ \hat{m}_t &= \frac{m_t}{1-\beta_1^t}, & \hat{v}_t &= \frac{v_t}{1-\beta_2^t}, & \theta_t &= \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda \theta_{t-1} \right). \end{aligned}$$

The last $\lambda\theta$ term is **AdamW**’s decoupled weight decay. Everything is element-wise ($\odot$, $\sqrt{\cdot}$, division all act entrywise), so Adam is *blind to matrix structure*: permuting the entries of a weight matrix would not change what Adam does to them.

Geometry. Adam is (approximately) steepest descent in a *diagonal* metric — it solves $\min_{\delta} \langle g, \delta \rangle + \frac{1}{2\eta} \|\delta\|_{\text{diag}(\sqrt{v})}^2$, giving $\delta \propto -g/\sqrt{v}$. Each coordinate is rescaled by its own history; the *correlations* between coordinates (the matrix’s singular structure) are ignored.

2 Muon — orthogonalize the momentum

MomentUm Orthogonalized by Newton-schulz

Muon keeps *only* momentum (one buffer), then — before stepping — replaces that momentum matrix by the *nearest orthogonal matrix*. For a 2-D weight W with momentum M_t :

$$M_t = \beta M_{t-1} + g_t, \quad O_t = \text{msign}(M_t) \approx UV^\top \text{ where } M_t = U\Sigma V^\top, \quad \theta_t = \theta_{t-1} - \eta (\gamma O_t + \lambda \theta_{t-1}).$$

$\text{msign}(M) = UV^\top$ is the *orthogonal polar factor* — it keeps M ’s singular *directions* U, V but sets *every singular value* to 1. Equivalently it is the closest (semi-)orthogonal matrix to M :

$$UV^\top = \arg \min_{O: O^\top O = I} \|O - M\|_F.$$

Computing it by SVD would be too slow, so Muon uses a **Newton–Schulz** quintic iteration (a few matmuls, run in bf16):

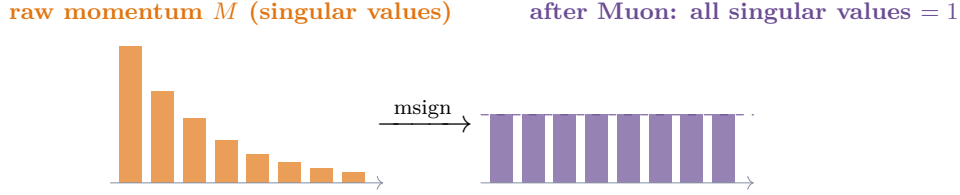
$$X_0 = \frac{M}{\|M\|_F}, \quad X_{k+1} = a X_k + b (X_k X_k^\top) X_k + c (X_k X_k^\top)^2 X_k, \quad (a, b, c) = (3.4445, -4.7750, 2.0315),$$

which drives all singular values toward 1 in ~ 5 steps. γ is a scale that RMS-matches Muon’s update to Adam’s so learning rates transfer (e.g. $\gamma \sim 0.2 \sqrt{\max(n, m)}$ for an $n \times m$ matrix).

Geometry. Muon is steepest descent in the *spectral* (operator) norm: $O_t = \arg \min_{\delta} \langle M_t, \delta \rangle$ s.t. $\|\delta\|_2 \leq 1 \Rightarrow \delta = UV^\top$. Where Adam bounds the update *coordinate-wise*, Muon bounds its largest *singular value* — it asks “what is the best-conditioned matrix step of unit operator norm?”

3 The picture: what orthogonalization does to the spectrum

Momentum of a transformer weight matrix is typically *spectrally skewed* — a few large singular values carry most of the energy. Adam’s element-wise scaling does nothing about this, so the update is dominated by a handful of directions. Muon flattens the spectrum to all-ones, so *every* direction gets an equal step:



Why this helps. Equal-magnitude steps in all singular directions means the update uses the *full rank* of the weight, instead of repeatedly pushing the one or two already-dominant directions. It is better conditioned, so each step makes more balanced progress — empirically faster convergence per step (Muon set nanoGPT speed records) and steadier loss curves at scale.

4 Practicalities (and why V4 mixes the two)

Scope — matrices only. Muon’s orthogonalization only makes sense for 2-D hidden weight matrices. Embeddings, the unembedding/LM head, and 1-D parameters (RMSNorm scales, biases) have no useful operator structure, so they stay on **AdamW**. This is exactly DeepSeek-V4’s split: *Muon for the matrices, AdamW for embeddings, the prediction head, and RMSNorm weights.*

Half the optimizer memory. Muon stores only momentum M ; Adam stores both m and v . At trillion-param scale, dropping the second-moment buffer is a real systems win (roughly halves optimizer state for the matrices).

Extra compute, cheap. The Newton–Schulz iterations add a few $d \times d$ matmuls per matrix per step — negligible next to forward/backward, doable in low precision and shardable across data-parallel ranks.

The scaling caveat (and a DeepSeek deletion). Muon’s aggressive, full-rank updates can inflate attention QK logits. Moonshot’s Kimi K2 scaled Muon only by adding **QK-Clip** / “**Muon-Clip**” to cap those logits. **DeepSeek-V4 dropped QK-Clip** — RMSNorm applied to the queries and compressed-KV entries already bounds the logits, so the stabilizer was unnecessary. (Same delete-the-crutch instinct as the rest of their stack.)

5 The third reference point: Shampoo

full Kronecker curvature

Adam estimates curvature *diagonally* (one scale per coordinate). Muon estimates no magnitudes at all — it keeps directions and *assumes* the rest is isotropic. **Shampoo** sits at the far end: it actually *measures* the anisotropy, via a Kronecker-factored approximation of full-matrix second-order information.

For a weight W with gradient $G \in \mathbb{R}^{m \times n}$, Shampoo accumulates two preconditioners — one for the *row* space, one for the *column* space — and steps with their inverse fourth-roots:

$$L = \sum_t G_t G_t^\top \in \mathbb{R}^{m \times m}, \quad R = \sum_t G_t^\top G_t \in \mathbb{R}^{n \times n}, \quad W \leftarrow W - \eta L^{-1/4} G R^{-1/4}.$$

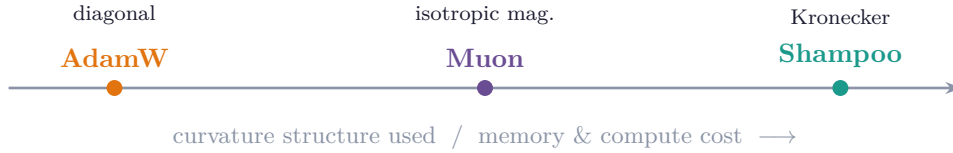
$L^{1/2} \otimes R^{1/2}$ is a Kronecker stand-in for the otherwise $mn \times mn$ full-matrix AdaGrad preconditioner, so each direction is rescaled by its *measured* curvature rather than by an assumption.

Muon is single-step Shampoo. Drop the accumulation — use only the current step’s factors $L = GG^\top$, $R = G^\top G$ — and the update collapses *exactly* to orthogonalization. With $G = U\Sigma V^\top$ we have $L^{-1/4} = U\Sigma^{-1/2}U^\top$ and $R^{-1/4} = V\Sigma^{-1/2}V^\top$, so

$$L^{-1/4}GR^{-1/4} = U\Sigma^{-1/2}U^\top \cdot U\Sigma V^\top \cdot V\Sigma^{-1/2}V^\top = U\Sigma^{-1/2}\Sigma\Sigma^{-1/2}V^\top = UV^\top = \text{msign}(G).$$

The Σ ’s cancel *exactly* — that cancellation *is* the spectrum-flattening. So Muon is Shampoo with **no memory**: where Shampoo accumulates curvature over many steps to estimate the true anisotropy, Muon uses the instantaneous factors, which cancel to pure isotropy. The “bias toward equal directions” is precisely what you get by refusing to accumulate.

The cost of measuring vs. assuming. Shampoo’s better estimate isn’t free: it stores $m^2 + n^2$ preconditioner entries per matrix and periodically computes their inverse fourth-roots (eigendecompositions / Newton iterations on $m \times m$ and $n \times n$). Muon keeps one $m \times n$ momentum buffer and a handful of matmuls. The three optimizers fall on a clean *cost vs. curvature-fidelity* line:



In practice Muon often *matches or beats* well-tuned Shampoo at a fraction of the memory and compute — the accumulated curvature estimate can be noisy or stale, and Muon’s instantaneous isotropy bias doubles as a cheap regularizer. That trade — nearly Shampoo’s conditioning, at one momentum buffer — is why DeepSeek (and others) reach for Muon rather than Shampoo at scale.

6 Side-by-side

	AdamW	Muon	Shampoo
Curvature used	diagonal (per coord.)	directions only (isotropic)	full Kronecker (row & col)
State per matrix	m, v (2 buffers)	M (1 buffer)	$L_{m \times m}, R_{n \times n}$
Update rule	$-\eta \hat{m} / (\sqrt{\hat{v}} + \epsilon)$	$-\eta UV^\top$	$-\eta L^{-1/4}GR^{-1/4}$
Implicit norm	per-coordinate RMS	spectral (operator)	estimated curvature
Effect on spectrum	leaves it skewed	flattens to 1	rescales by measured curvature
Extra compute	none	Newton–Schulz (~ 5 iters)	inverse 4th-roots of L, R
Memory	$2 \times$ params	$1 \times$ params	$+ m^2 + n^2$ per matrix

(In V4: **Muon** for all hidden weight matrices, **AdamW** for embeddings, the prediction head, and RMSNorm weights. Shampoo is included as the conceptual far end, not part of V4.)

Bottom line. Adam is a safe, structure-agnostic per-coordinate preconditioner. Muon adds the one piece of structure that matters for a weight matrix — its singular spectrum — by orthogonalizing the momentum, buying better-conditioned updates, faster convergence, and half the optimizer state,

at the price of a cheap matmul iteration and applicability to matrices only. V4 uses each where it fits.

Muon: Jordan et al. (2024); spectral-norm-steepest-descent framing: Bernstein & Newhouse, “modular norm.” Shampoo: Gupta, Anil & Koren (2018); Distributed Shampoo: Anil et al. (2020); the Muon-as-instantaneous-Shampoo identity is standard folklore in that line of work. Kimi K2 / MuonClip: Moonshot (2025). The V4 Muon configuration (which parameters, dropping QK-Clip) is from third-party summaries of the DeepSeek-V4 technical report (Apr–May 2026) and is reported as published.